

Lab 1 GE05021: Continuing with linux and starting with OpenFOAM

Modelling urban flows

Learning Objectives:

After this class, you will be able to:

- How to run openFOAM in your machines
- Install and run paraView

Activities:

This class is hands-on, so you will need to follow my steps with your own computer, so no activities will be carried out, but I will still ask you questions....

What we will learn might be very new to some, and not so new to others, if you have any doubt it is very important to solve it asap, so you don't get lost

Activities:

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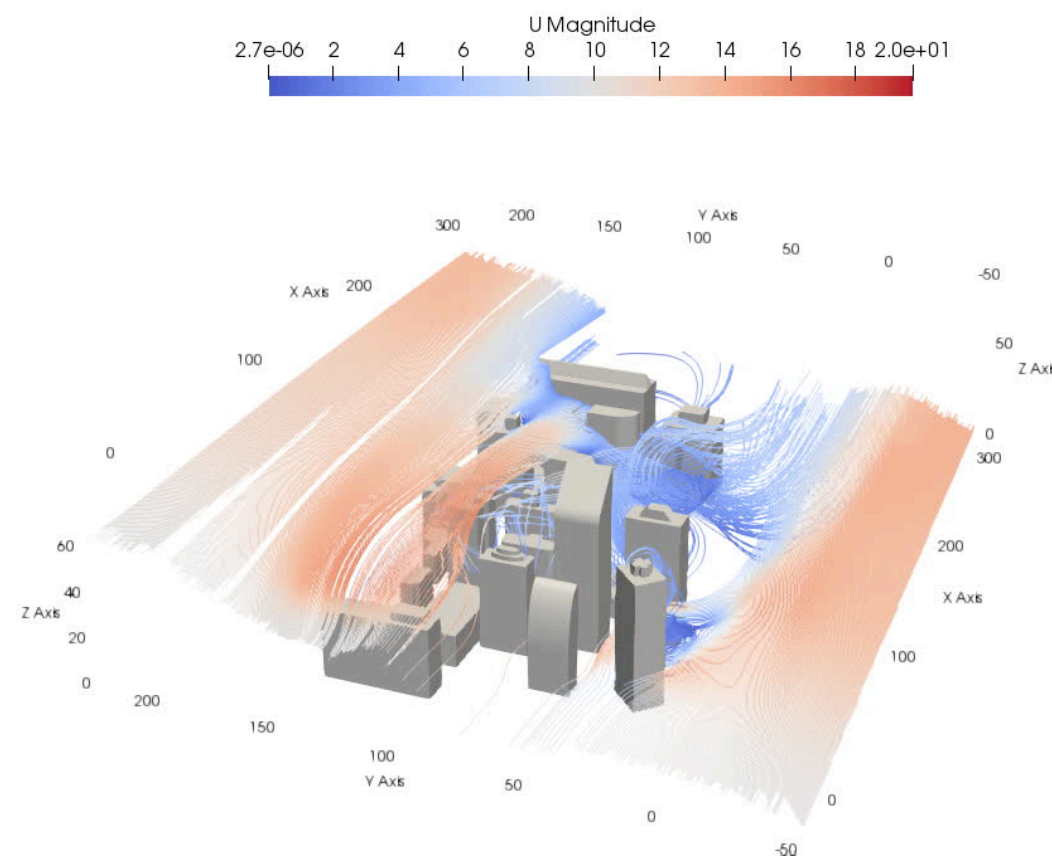
Remember you will use what I show you today, to run your own case for the assignment!!!

Outline of the class:

1. What is openFOAM?
2. openFOAM test case structure
3. How to run openFOAM test case
4. Visualise the results with paraView

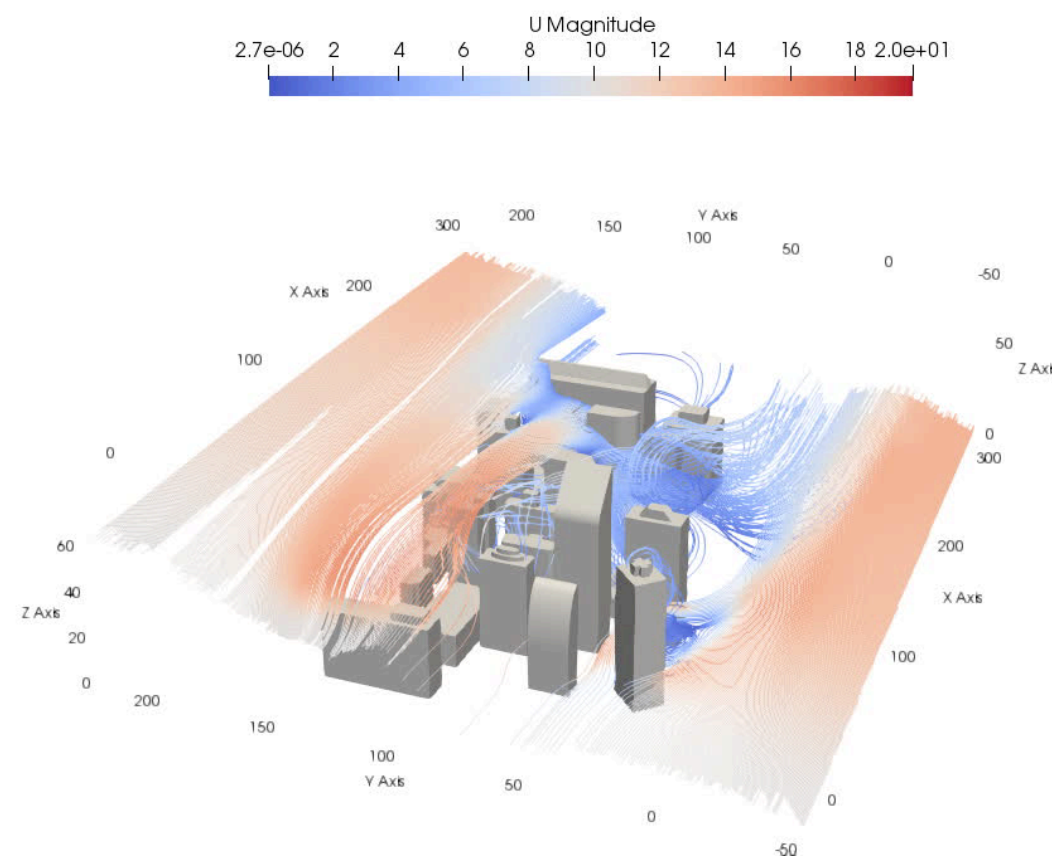
What is openFOAM?

- It is a collection of about 250 applications built from a collection of 100 software libraries (modules)
- Each application performs a specific task within a CFD workflow: snappyHexMesh will generate a mesh for complex geometry for example a car, while simpleFoam will then simulate steady-state, turbulent, incompressible flow around that car.



What is openFOAM?

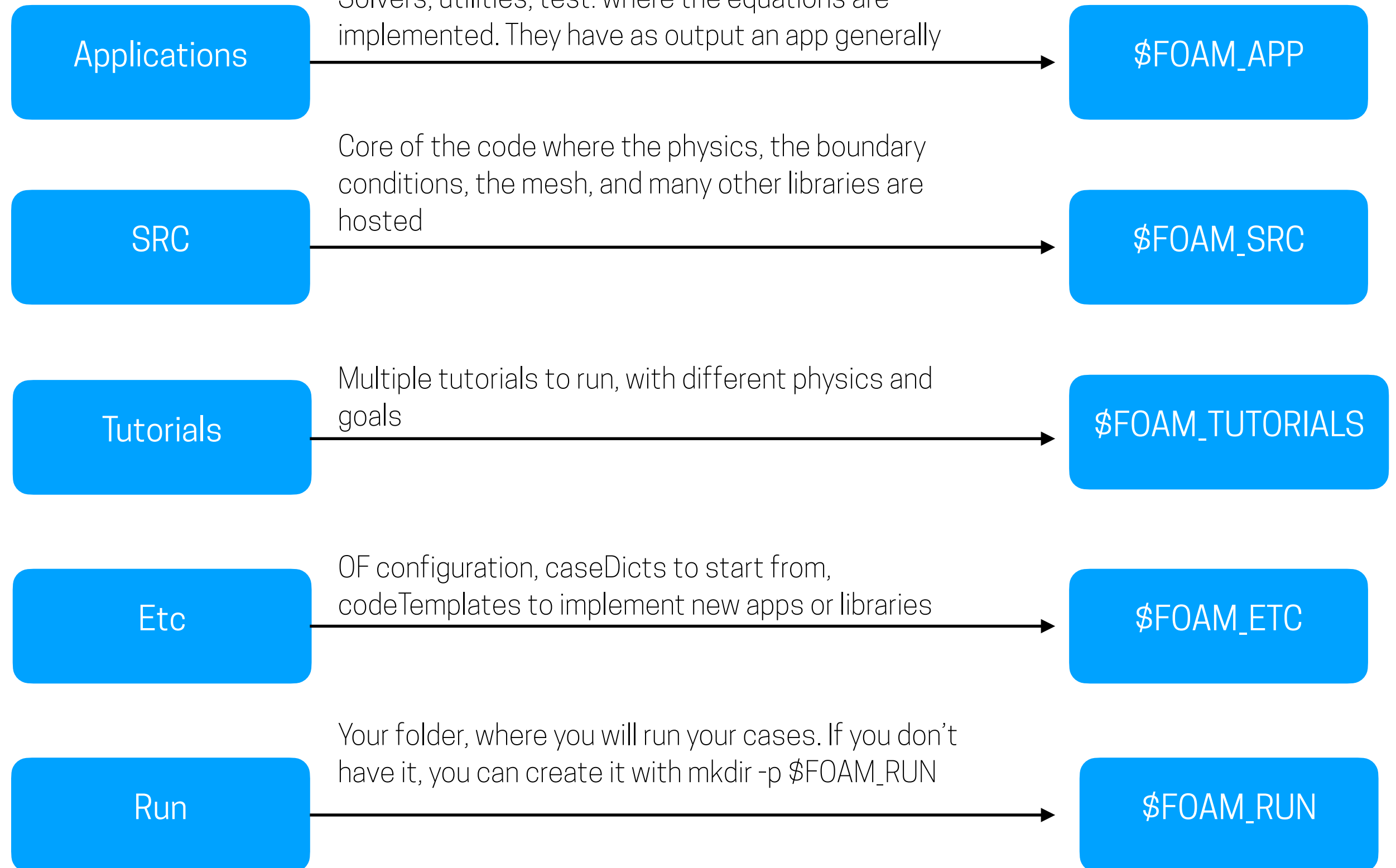
- It is a collection of about 250 applications built from a collection of 100 software libraries (modules)
- Each application performs a specific task within a CFD workflow: snappyHexMesh will generate a mesh for complex geometry for example a car, while simpleFoam will then simulate steady-state, turbulent, incompressible flow around that car.



openFOAM test case structure

Folders tree code:

Symbolic path:



openFOAM test case structure

- Folders tree in case:

```
cp -r $FOAM_TUTORIALS/incompressible/simpleFoam/windAroundBuildings $FOAM_RUN/
```

```
gsclara@foote:/opt/OpenFOAM-v2412/tutorials/incompressible/simpleFoam/windAroundBuildings$ ls -la
total 36
drwxr-xr-x  5 root root 4096 Dec 20 2024 .
drwxr-xr-x 19 root root 4096 Dec 20 2024 ..
drwxr-xr-x  2 root root 4096 Dec 20 2024 0.orig
-rwxr-xr-x  1 root root  408 Dec 20 2024 Allclean
-rwxr-xr-x  1 root root  387 Dec 20 2024 Allrun
-rwxr-xr-x  1 root root  516 Dec 20 2024 Allrun-parallel
-rwxr-xr-x  1 root root  555 Dec 20 2024 Allrun.pre
drwxr-xr-x  2 root root 4096 Dec 20 2024 constant
drwxr-xr-x  2 root root 4096 Dec 20 2024 system
```

openFOAM test case structure

- Folders tree in case:

```
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```

```
gsclara@foote:/opt/OpenFOAM-v2412/tutorials/incompressible/simpleFoam/windAroundBuildings$ ls -la
total 36
drwxr-xr-x  5 root root 4096 Dec 20 2024 .
drwxr-xr-x 19 root root 4096 Dec 20 2024 ..
drwxr-xr-x  2 root root 4096 Dec 20 2024 0.orig
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-rwxr-xr-x  1 root root  555 Dec 20 2024 Allrun.pre
drwxr-xr-x  2 root root 4096 Dec 20 2024 constant
drwxr-xr-x  2 root root 4096 Dec 20 2024 system
```



openFOAM test case structure

- Allrun:

```
#!/bin/sh
cd "${0%/*}" || exit                # Run from this directory
. ${WM_PROJECT_DIR:?}/bin/tools/RunFunctions  # Tutorial run functions
#-----

./Allrun.pre

restore0Dir

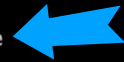
runApplication $(getApplication)

#-----
```

openFOAM test case structure

- Allrun:

```
#!/bin/sh
cd "${0%/*}" || exit
. ${WM_PROJECT_DIR:?}/bin/tools/RunFunctions
#-----
./Allrun.pre
restore0Dir
runApplication $(getApplication)
```



```
#!/bin/sh
cd "${0%/*}" || exit
. ${WM_PROJECT_DIR:?}/bin/tools/RunFunctions
#-----

mkdir -p constant/triSurface

cp -f \
    "$FOAM_TUTORIALS"/resources/geometry/buildings.obj.gz \
    constant/triSurface/

runApplication surfaceFeatureExtract

runApplication blockMesh

runApplication snappyHexMesh -overwrite

#-----
```

openFOAM test case structure

- Allrun:

```
#!/bin/sh
cd "${0%/*}" || exit                # Run from this directory
. ${WM_PROJECT_DIR:?}/bin/tools/RunFunctions  # Tutorial run functions
#-----

./Allrun.pre

restore0Dir

runApplication $(getApplication)

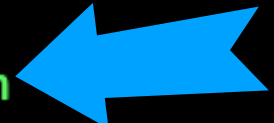
#-----
```

openFOAM test case structure

- Folders tree in case:

```
cp -r $FOAM_TUTORIALS/incompressible/simpleFoam/windAroundBuildings $FOAM_RUN/
```

```
gsclara@foote:/opt/OpenFOAM-v2412/tutorials/incompressible/simpleFoam/windAroundBuildings$ ls -la
total 36
drwxr-xr-x  5 root root 4096 Dec 20 2024 .
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drwxr-xr-x  2 root root 4096 Dec 20 2024 constant
drwxr-xr-x  2 root root 4096 Dec 20 2024 system
```



openFOAM test case structure

- Allclean:

```
#!/bin/sh
cd "${0%/*}" || exit                # Run from this directory
. ${WM_PROJECT_DIR:?}/bin/tools/CleanFunctions    # Tutorial clean functions
#-----

cleanCase0

rm -rf constant/extendedFeatureEdgeMesh
rm -rf constant/triSurface

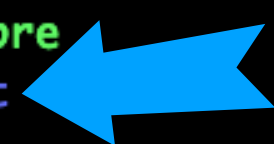
#-----
```

openFOAM test case structure

- Folders tree in case:

```
cp -r $FOAM_TUTORIALS/incompressible/simpleFoam/windAroundBuildings $FOAM_RUN/
```

```
gsclara@foote:/opt/OpenFOAM-v2412/tutorials/incompressible/simpleFoam/windAroundBuildings$ ls -la
total 36
drwxr-xr-x  5 root root 4096 Dec 20 2024 .
drwxr-xr-x 19 root root 4096 Dec 20 2024 ..
drwxr-xr-x  2 root root 4096 Dec 20 2024 0.orig
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-rwxr-xr-x  1 root root  516 Dec 20 2024 Allrun-parallel
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drwxr-xr-x  2 root root 4096 Dec 20 2024 constant
drwxr-xr-x  2 root root 4096 Dec 20 2024 system
```



openFOAM test case structure

- Constant:

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/constant$ ls -la
total 20
drwxr-xr-x 3 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 15:43 ..
-rw-r--r-- 1 gsclara gsclara  898 Mar  4 15:43 transportProperties
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 15:43 triSurface
-rw-r--r-- 1 gsclara gsclara  960 Mar  4 15:43 turbulenceProperties
```

```
/*-----*-- C++ -----*--\
|=====|
|  \  /  F i e l d      | OpenFOAM: The Open Source CFD Toolbox
|  \  /  O p e r a t i o n | Version:  v2412
|  \  /  A n d           | Website:  www.openfoam.com
|  \  /  M a n i p u l a t i o n |
|=====|
\*-----*--\

FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       transportProperties;
}

// *****

transportModel  Newtonian;

nu              1.5e-05;

// *****
```

openFOAM test case structure

- Constant:

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/constant$ ls -la
total 20
drwxr-xr-x 3 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 15:43 ..
-rw-r--r-- 1 gsclara gsclara  898 Mar  4 15:43 transportProperties
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 15:43 triSurface
-rw-r--r-- 1 gsclara gsclara  960 Mar  4 15:43 turbulenceProperties
```

```
/*-----* C++ *-----*/
|=====|
| \ \ /  F i e l d      | OpenFOAM: The Open Source CFD Toolbox
| \ \ /  O peration     | Version:  v2412
| \ \ /  A nd           | Website:  www.openfoam.com
|  \ \  M anipulation   |
|-----*/

FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       turbulenceProperties;
}
// *****

simulationType    RAS;

RAS
{
    RASModel       kEpsilon;

    turbulence     on;

    printCoeffs    on;
}

// *****
```

openFOAM test case structure

- Constant:

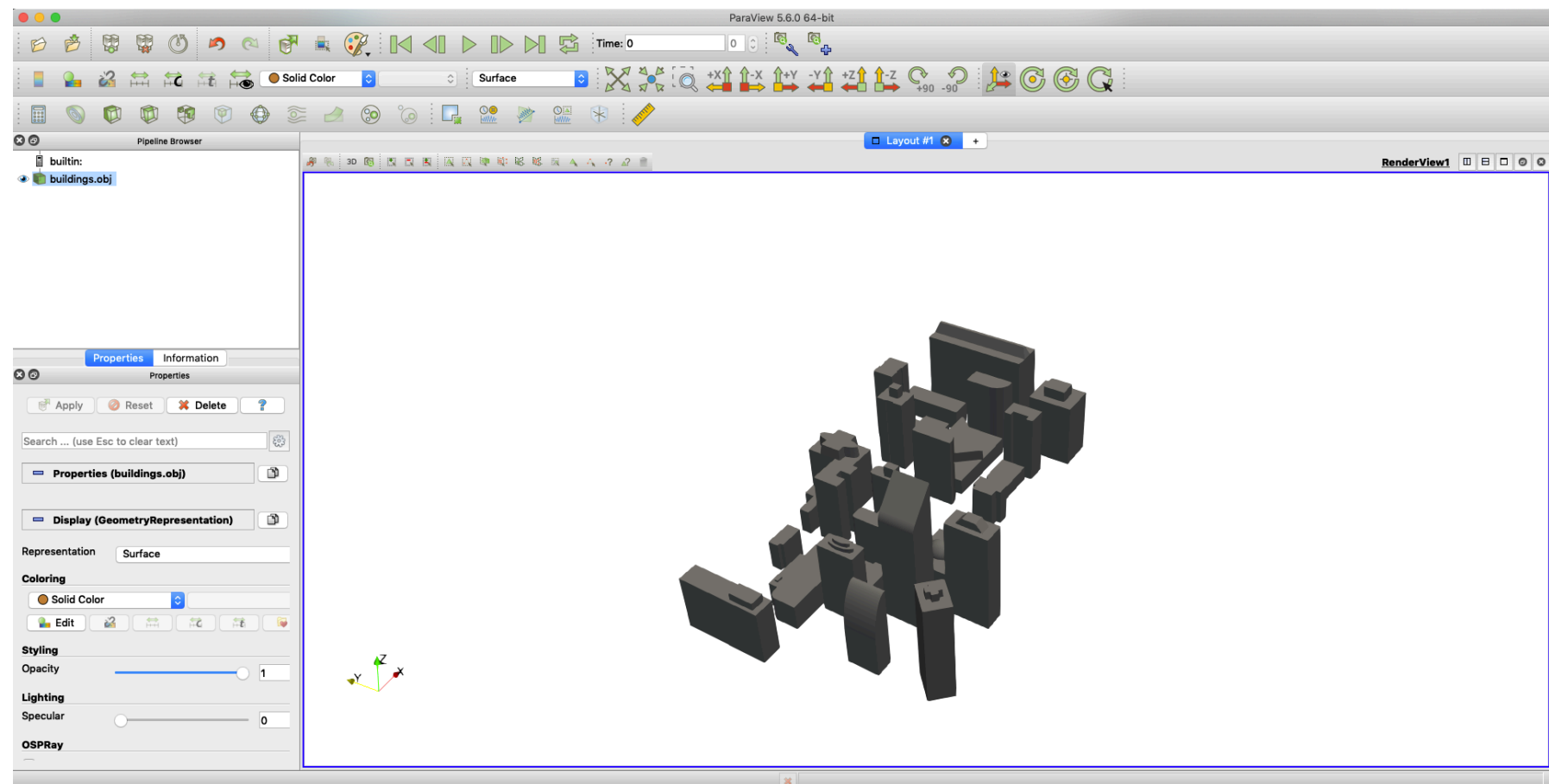
Geometry —>

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/constant$ ls -la
total 20
drwxr-xr-x 3 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 15:43 ..
-rw-r--r-- 1 gsclara gsclara  898 Mar  4 15:43 transportProperties
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 15:43 triSurface
-rw-r--r-- 1 gsclara gsclara  960 Mar  4 15:43 turbulenceProperties
```

Name	Date Modified
 buildings.obj.gz	Yesterday at 17:11

gzip -d buildings.obj.gz

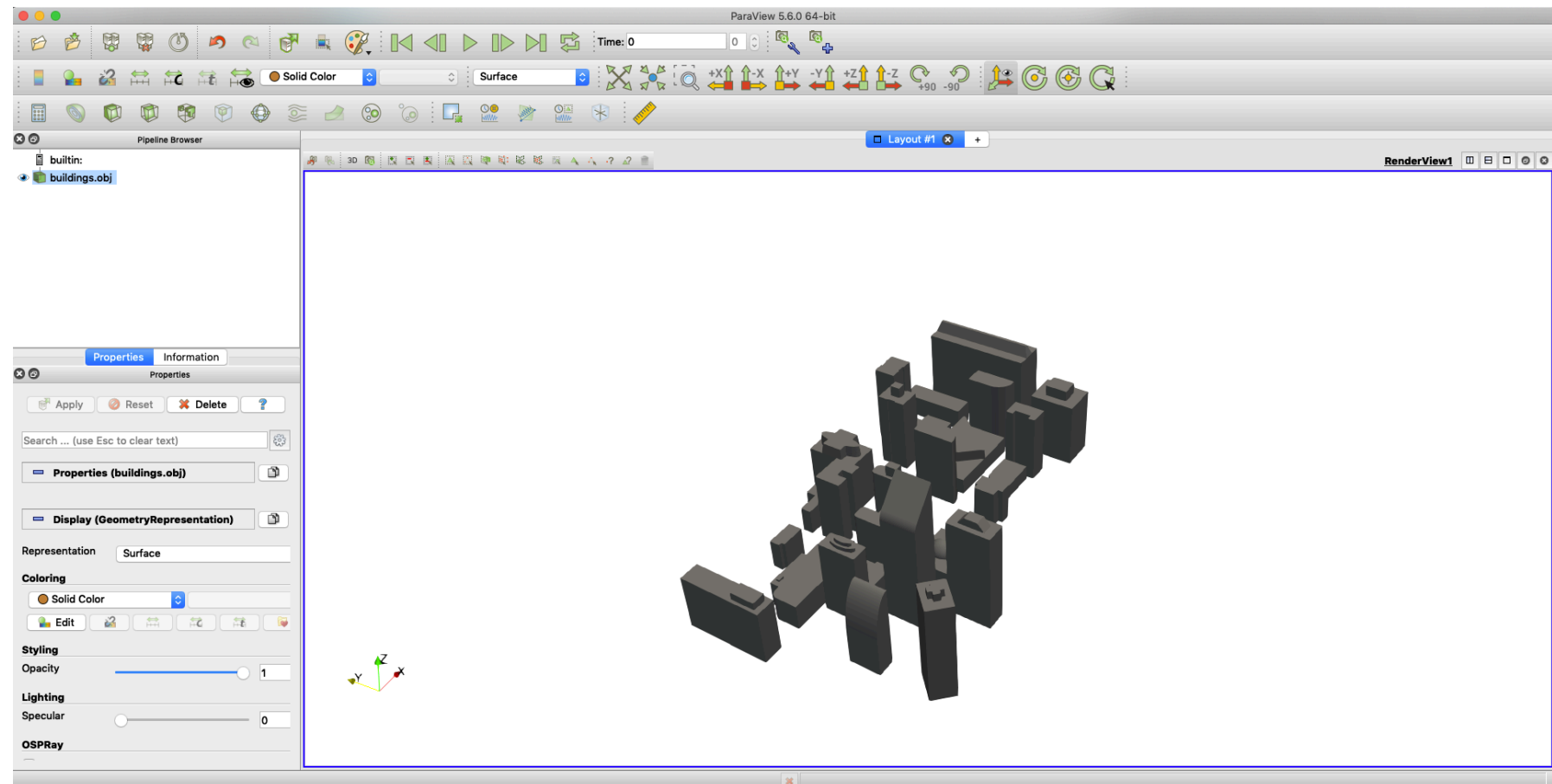
Open buildings.obj with Paraview



Let's install paraView now!

Go to: <https://www.paraview.org/download/>

To visualise an openFOAM case you always need a dummy file with any name and .foam extension, like windAroundBuildings.foam



openFOAM test case structure

- Folders tree in case:

```
cp -r $FOAM_TUTORIALS/incompressible/simpleFoam/windAroundBuildings $FOAM_RUN/
```

```
gsclara@foote:/opt/OpenFOAM-v2412/tutorials/incompressible/simpleFoam/windAroundBuildings$ ls -la
total 36
drwxr-xr-x  5 root root 4096 Dec 20 2024 .
drwxr-xr-x 19 root root 4096 Dec 20 2024 ..
drwxr-xr-x  2 root root 4096 Dec 20 2024 0.orig ←
-rwxr-xr-x  1 root root  408 Dec 20 2024 Allclean
-rwxr-xr-x  1 root root  387 Dec 20 2024 Allrun
-rwxr-xr-x  1 root root  516 Dec 20 2024 Allrun-parallel
-rwxr-xr-x  1 root root  555 Dec 20 2024 Allrun.pre
drwxr-xr-x  2 root root 4096 Dec 20 2024 constant
drwxr-xr-x  2 root root 4096 Dec 20 2024 system
```

openFOAM test case structure

- 0: contains the initial and boundary conditions for the fields to be modelled

The basic boundary conditions are:

- fixedValue: value of field is specified
- fixedGradient: value of the normal field gradient is specified
- zeroGradient: normal field gradient is null (0)
- calculated: patch field value is calculated from other patches
- mixed: mixed fixedValue/fixedGradient condition depending on a valueFraction ($0(\text{refGradient}) < \text{vf} < 1(\text{refValue})$)
- directionMixed: mixed conditions with tensorial valueFraction to allow different conditions in normal and tangential direction of a vector patch field (fixedValue in tangential, zeroGradient in normal direction)

simpleFoam -listScalarBCs -listVectorBCs

foamInfo totalPressure

openFOAM test case structure

- 0: U - velocity initial and boundary values:

Name	Date Modified
epsilon	Yesterday at 17:11
k	Yesterday at 17:11
nut	Yesterday at 17:11
p	Yesterday at 17:11
U	Yesterday at 17:11

```
/*-----* C++ *-----*/
|=====|
| \ \ / \ F i e l d | OpenFOAM: The Open Source CFD Toolbox |
| \ \ / \ O p e r a t i o n | Version: v2412 |
| \ \ / \ A n d | Website: www.openfoam.com |
| \ \ / \ M a n i p u l a t i o n |
|=====|
/*-----*/

FoamFile
{
    version      2.0;
    format       ascii;
    class        volVectorField;
    object       U;
}

// *****

Uinlet          (10 0 0);

dimensions      [0 1 -1 0 0 0];

internalField   uniform (0 0 0);

boundaryField
{
    inlet
    {
        type      fixedValue;
        value      uniform $Uinlet;
    }

    outlet
    {
        type      pressureInletOutletVelocity;
        value      uniform (0 0 0);
    }

    wall
    {
        type      noSlip;
    }

    #includeEtc "caseDicts/setConstraintTypes"
}
}
```

openFOAM test case structure

- 0: U - velocity initial and boundary values:

Name	Date Modified
epsilon	Yesterday at 17:11
k	Yesterday at 17:11
nut	Yesterday at 17:11
p	Yesterday at 17:11
U	Yesterday at 17:11

```

/*-----* C++ *-----*/
|=====|
| \ / | F i e l d | OpenFOAM: The Open Source CFD Toolbox |
| \ / | O p e r a t i o n | Version: v2412 |
| \ / | A n d | Website: www.openfoam.com |
| \ / | M a n i p u l a t i o n |
|=====|
FoamFile
{
    version      2.0;
    format       ascii;
    class        volVectorField;
    object       U;
}
// *****

```

No.	Property	SI unit	USCS unit
1	Mass	kilogram (kg)	pound-mass (lbm)
2	Length	metre (m)	foot (ft)
3	Time	second (s)	second (s)
4	Temperature	Kelvin (K)	degree Rankine (°R)
5	Quantity	mole (mol)	mole (mol)
6	Current	ampere (A)	ampere (A)
7	Luminous intensity	candela (cd)	candela (cd)

```

{
    type          noSlip;
}

#includeEtc "caseDicts/setConstraintTypes"
}

```

openFOAM test case structure

- 0: p - pressure initial and boundary values:

Name	Date Modified
epsilon	Yesterday at 17:11
k	Yesterday at 17:11
nut	Yesterday at 17:11
p	Yesterday at 17:11
U	Yesterday at 17:11

```
/*-----* C++ *-----*/
|=====|
| \ /   | F i e l d       | OpenFOAM: The Open Source CFD Toolbox |
| \ /   | O peration      | Version:  v2412                      |
| \ /   | A nd            | Website:   www.openfoam.com           |
| \ /   | M anipulation   |                                     |
|=====|
/*-----*/
FoamFile
{
    version      2.0;
    format       ascii;
    class        volScalarField;
    object       p;
}
// *****

dimensions      [0 2 -2 0 0 0 0];

internalField   uniform 0;

boundaryField
{
    inlet
    {
        type      zeroGradient;
    }

    outlet
    {
        type      totalPressure;
        p0        uniform 0;
    }

    wall
    {
        type      zeroGradient;
    }

    #includeEtc "caseDicts/setConstraintTypes"
}
}
```

openFOAM test case structure

- 0: k - tke initial and boundary values:

Name	Date Modified
epsilon	Yesterday at 17:11
k	Yesterday at 17:11
nut	Yesterday at 17:11
p	Yesterday at 17:11
U	Yesterday at 17:11

```
/*----- C++ -----*/
|=====|
| \ \ / Field | OpenFOAM: The Open Source CFD Toolbox |
| \ \ / Operation | Version: v2412 |
| \ \ / And | Website: www.openfoam.com |
| \ \ / Manipulation |
|=====|
FoamFile
{
    version      2.0;
    format       ascii;
    class        volScalarField;
    object       k;
}
// *****

kInlet          1.5;    // approx k = 1.5*(I*U)^2 ; I = 0.1

dimensions      [0 2 -2 0 0 0];

internalField   uniform $kInlet;

boundaryField
{
    inlet
    {
        type      fixedValue;
        value      uniform $kInlet;
    }

    outlet
    {
        type      inletOutlet;
        inletValue uniform $kInlet;
        value      uniform $kInlet;
    }

    wall
    {
        type      kqRWallFunction;
        value      uniform $kInlet;
    }

    #includeEtc "caseDicts/setConstraintTypes"
}
```

openFOAM test case structure

- 0: ϵ - dissipation initial and boundary values:

Name		Date Modified
epsilon		Yesterday at 17:11
k		Yesterday at 17:11
nut		Yesterday at 17:11
p		Yesterday at 17:11
U		Yesterday at 17:11

```
/*-----* C++ *-----*/
|=====|
| \ / | F i e l d | OpenFOAM: The Open Source CFD Toolbox |
| \ / | O p e r a t i o n | Version: v2412 |
| \ / | A n d | Website: www.openfoam.com |
| \ / | M a n i p u l a t i o n |
|=====|
/*-----*/

FoamFile
{
    version      2.0;
    format       ascii;
    class        volScalarField;
    object       epsilon;
}

// *****

epsilonInlet  0.03; // Cmu^0.75 * k^1.5 / L ; L =10

dimensions    [0 2 -3 0 0 0];

internalField uniform $epsilonInlet;

boundaryField
{
    inlet
    {
        type      fixedValue;
        value      uniform $epsilonInlet;
    }

    outlet
    {
        type      inletOutlet;
        inletValue uniform $epsilonInlet;
        value      uniform $epsilonInlet;
    }

    wall
    {
        type      epsilonWallFunction;
        value      uniform $epsilonInlet;
    }

    #includeEtc "caseDicts/setConstraintTypes"
}
```

openFOAM test case structure

- 0: ν_t - turbulent viscosity initial and boundary values:

Name	Date Modified
epsilon	Yesterday at 17:11
k	Yesterday at 17:11
nut	Yesterday at 17:11
p	Yesterday at 17:11
U	Yesterday at 17:11

```
/*-----* C++ -*-----*/
|=====|
| \\  /  | F i e l d      | OpenFOAM: The Open Source CFD Toolbox |
| \\  /  | O peration     | Version: v2412                        |
| \\  /  | A nd           | Website: www.openfoam.com              |
| \\  /  | M anipulation  |                                     |
|=====|
FoamFile
{
    version      2.0;
    format       ascii;
    class        volScalarField;
    object       nut;
}
// *****

dimensions      [0 2 -1 0 0 0 0];

internalField   uniform 0;

boundaryField
{
    inlet
    {
        type      calculated;
        value      uniform 0;
    }

    outlet
    {
        type      calculated;
        value      uniform 0;
    }

    wall
    {
        type      nutkWallFunction;
        value      uniform 0;
    }
}

#includeEtc "caseDicts/setConstraintTypes"
}
```

openFOAM test case structure

- Folders tree in case:

```
cp -r $FOAM_TUTORIALS/incompressible/simpleFoam/windAroundBuildings $FOAM_RUN/
```

```
gsclara@foote:/opt/OpenFOAM-v2412/tutorials/incompressible/simpleFoam/windAroundBuildings$ ls -la
total 36
drwxr-xr-x  5 root root 4096 Dec 20 2024 .
drwxr-xr-x 19 root root 4096 Dec 20 2024 ..
drwxr-xr-x  2 root root 4096 Dec 20 2024 0.orig
-rwxr-xr-x  1 root root  408 Dec 20 2024 Allclean
-rwxr-xr-x  1 root root  387 Dec 20 2024 Allrun
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-rwxr-xr-x  1 root root  555 Dec 20 2024 Allrun.pre
drwxr-xr-x  2 root root 4096 Dec 20 2024 constant
drwxr-xr-x  2 root root 4096 Dec 20 2024 system
```



openFOAM test case structure

- System - BlockMeshDict:

```
/*-----* C++ *-----*/
|=====|
| \ \ / / F i e l d | OpenFOAM: The Open Source CFD Toolbox |
| \ \ / / O p e r a t i o n | Version: v2412 |
| \ \ / / A n d | Website: www.openfoam.com |
| \ \ / / M a n i p u l a t i o n |
|=====|
FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       blockMeshDict;
}

// *****

domain
{
    xMin  -20; // L = 350
    xMax  330;
    yMin  -50; // L = 280
    yMax  230;
    zMin   0;
    zMax  140;
    xCells 25;
    yCells 20;
    zCells 10;
}

scale 1;

// Note: uses older '.' syntax (to test compatibility)
vertices
(
    ($:domain.xMin $:domain.yMin $:domain.zMin)
    ($:domain.xMax $:domain.yMin $:domain.zMin)
    ($:domain.xMax $:domain.yMax $:domain.zMin)
    ($:domain.xMin $:domain.yMax $:domain.zMin)

    ($:domain.xMin $:domain.yMin $:domain.zMax)
    ($:domain.xMax $:domain.yMin $:domain.zMax)
    ($:domain.xMax $:domain.yMax $:domain.zMax)
    ($:domain.xMin $:domain.yMax $:domain.zMax)
);
```

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara  889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara  549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara  901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara  177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara  925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara  885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara  935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara  403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite
```

openFOAM test case structure

- System - BlockMeshDict:

```
boundary
(
    inlet
    {
        type patch;
        faces
        (
            (0 3 7 4)
        );
    }

    outlet
    {
        type patch;
        faces
        (
            (1 5 6 2)
        );
    }

    ground
    {
        type wall;
        faces
        (
            (0 1 2 3)
        );
    }

    frontAndBack
    {
        type symmetry;
        faces
        (
            (0 4 5 1)
            (3 2 6 7)
            (4 7 6 5)
        );
    }
);
```

```
blocks
(
    hex (0 1 2 3 4 5 6 7)
    (
        $:domain.xCells
        $:domain.yCells
        $:domain.zCells
    )
    simpleGrading (1 1 1)
);
```

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara  889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara  549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara  901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara  177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara  925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara  885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara  935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara  403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite
```

openFOAM test case structure

- System - surfaceFeaturesDict:

```
/*-----*-- C++ -*-----*\
|=====|
| \ \ / / F i e l d      | OpenFOAM: The Open Source CFD Toolbox
| \ \ / / O p e r a t i o n | Version:  v2412
| \ \ / / A n d           | Website:  www.openfoam.com
| \ \ / / M a n i p u l a t i o n |
|=====|
\*-----*-- C++ -*-----*/

FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       surfaceFeatureExtractDict;
}

// *****

buildings.obj
{
    #includeEtc "caseDicts/surface/surfaceFeatureExtractDict.cfg"
}

// *****
```

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara  889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara  549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara  901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara  177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara  925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara  885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara  935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara  403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite
```

```
/*-----*-- C++ -*-----*\
|=====|
| \ \ / / F i e l d      | OpenFOAM: The Open Source CFD Toolbox
| \ \ / / O p e r a t i o n | Version:  v2412
| \ \ / / A n d           | Website:  www.openfoam.com
| \ \ / / M a n i p u l a t i o n |
|=====|
\*-----*-- C++ -*-----*/

FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       surfaceFeatureExtractDict.cfg;
}

// *****

extractionMethod      extractFromSurface;

includedAngle         150;

geometricTestOnly     no;

subsetFeatures
{
    nonManifoldEdges  yes;
    openEdges         yes;
}

trimFeatures
{
    minElem           0;
    minLen             0;
}

writeObj              yes;

// *****
```

- System - SnappyHexMeshDict:

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara  889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara  549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara  901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara  177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara  925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara  885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara  935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara  403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite
```

```
snapControls
{
    explicitFeatureSnap    true;
    implicitFeatureSnap    false;
}
```



openFOAM test case structure

- System - meshQualityDict:

```
----- C++ -----
| \ \ / F i e l d | OpenFOAM: The Open Source CFD Toolbox |
| \ \ / O p e r a t i o n | Version: v2412 |
| \ \ / A n d | Website: www.openfoam.com |
| \ \ / M a n i p u l a t i o n |
|-----|

FoamFile
{
    version      2.0;
    format        ascii;
    class         dictionary;
    object        meshQualityDict.cfg;
}

// *****

maxNonOrtho 65;

maxBoundarySkewness 20;

maxInternalSkewness 4;

maxConcave 80;

// Minimum cell pyramid volume; case dependent
minVol 1e-13;

// 1e-15 (small positive) to enable tracking
// -1e+30 (large negative) for best layer insertion
minTetQuality 1e-15;

// if >0 : preserve single cells with all points on the surface if the
// resulting volume after snapping (by approximation) is larger than
// minVolCollapseRatio times old volume (i.e. not collapsed to flat cell).
// If <0 : delete always.
//minVolCollapseRatio 0.5;

minArea -1;

minTwist 0.02;

minDeterminant 0.001;

minFaceWeight 0.05;

minVolRatio 0.01;

minTriangleTwist -1;

nSmoothScale 4;

errorReduction 0.75;

relaxed
{
    maxNonOrtho 75;
}

// *****
```

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara 889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara 549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara 901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara 177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara 925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara 885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara 935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara 403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite
```

```
----- C++ -----
| \ \ / F i e l d | OpenFOAM: The Open Source CFD Toolbox |
| \ \ / O p e r a t i o n | Version: v2412 |
| \ \ / A n d | Website: www.openfoam.com |
| \ \ / M a n i p u l a t i o n |
|-----|

FoamFile
{
    version      2.0;
    format        ascii;
    class         dictionary;
    object        meshQualityDict;
}

// *****

#includeEtc "caseDicts/mesh/generation/meshQualityDict.cfg"

// *****
```

openFOAM test case structure

■ System - fvSchemes:

```
*-----* C++ *-----*
|=====|
| \ \ \ | F i e l d | OpenFOAM: The Open Source CFD Toolbox
| \ \ \ | O p e r a t i o n | Version: v2412
| \ \ \ | A n d | Website: www.openfoam.com
| \ \ \ | M a n i p u l a t i o n |
|=====|
FoamFile
{
  version      2.0;
  format       ascii;
  class        dictionary;
  object       fvSchemes;
}
// * * * * *

ddtSchemes
{
  default      steadyState;
}

gradSchemes
{
  default      Gauss linear;

  limited      cellLimited Gauss linear 1;
  grad(U)      $limited;
  grad(k)      $limited;
  grad(epsilon) $limited;
}

divSchemes
{
  default      none;

  div(phi,U)   bounded Gauss linearUpwind limited;

  turbulence    bounded Gauss limitedLinear 1;
  div(phi,k)    $turbulence;
  div(phi,epsilon) $turbulence;

  div((nuEff*dev2(T(grad(U)))) Gauss linear;
}

laplacianSchemes
{
  default      Gauss linear corrected;
}

interpolationSchemes
{
  default      linear;
}
```

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara  889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara  549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara  901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara  177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara  925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara  885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara  935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara  403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite
```

openFOAM test case structure

- System - fvSolution:

```

/*-----* C++ *-----*/
//
// \ /  F i e l d      | OpenFOAM: The Open Source CFD Toolbox
// \ /  O peration    | Version:  v2412
// \ /  A nd          | Website:  www.openfoam.com
// \ /  M anipulation |
//
foamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       fvSolution;
}
// *****

solvers
{
    p
    {
        solver      GAMG;
        smoother     GaussSeidel;
        tolerance    1e-6;
        relTol       0.1;
    }

    "(U|k|omega|epsilon)"
    {
        solver      smoothSolver;
        smoother     symGaussSeidel;
        tolerance    1e-6;
        relTol       0.1;
    }
}

SIMPLE
{
    residualControl
    {
        p          1e-4;
        U          1e-4;
        "(k|omega|epsilon)" 1e-4;
    }
    nNonOrthogonalCorrectors 0;
    pRefCell 0;
    pRefValue 0;
}

potentialFlow
{
    nNonOrthogonalCorrectors 10;
}

relaxationFactors
{
    fields
    {
        p          0.3;
    }
    equations
    {
        U          0.7;
        "(k|omega|epsilon)." 0.7;
    }
}

```

```

gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara  889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara  549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara  901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara  177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara  925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara  885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara  935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara  403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite

```

Successive overrelaxation:

- The convergence to a solution can be enhanced by utilizing successive overrelaxation
- We use an extrapolation procedure where the intermediate nodal variables $\phi_j^{(k+1)}$ are further advanced by a weighted average of the current values of $\phi_j^{(k+1)}$ with the previous values of $\phi_j^{(k)}$:

$$\overline{\phi_j}^{(k+1)} = (1 - \lambda)\phi_j^k + \lambda\phi_j^{(k+1)}$$

openFOAM test case structure

■ System - controlDict:

```
/*-----* C++ *-----*/
|=====|
| \ \ / / F i e l d | OpenFOAM: The Open Source CFD Toolbox |
| \ \ / / O p e r a t i o n | Version: v2412 |
| \ \ / / A n d | Website: www.openfoam.com |
| \ \ / / M a n i p u l a t i o n |
|=====|
FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       controlDict;
}
// *****

application      simpleFoam;

startFrom         latestTime;

startTime         0;

stopAt            endTime;

endTime           400;

deltaT            1;

writeControl      timeStep;

writeInterval     50;

purgeWrite        0;

writeFormat       binary;

writePrecision    6;

writeCompression off;

timeFormat        general;

timePrecision     6;

runTimeModifiable true;

functions
{
    #include "ensightWrite"
    #include "vtkWrite"

    // Demonstrate runTimePostProcessing
    #include "visualization"
    #include "profiling"
}
```

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/system$ ls -la
total 72
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:43 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
-rw-r--r-- 1 gsclara gsclara 2167 Mar  4 15:43 blockMeshDict
-rw-r--r-- 1 gsclara gsclara 1352 Mar  4 15:43 controlDict
-rw-r--r-- 1 gsclara gsclara  889 Mar  4 15:43 decomposeParDict
-rw-r--r-- 1 gsclara gsclara  549 Mar  4 15:43 ensightWrite
-rw-r--r-- 1 gsclara gsclara 1554 Mar  4 15:43 fvSchemes
-rw-r--r-- 1 gsclara gsclara 1602 Mar  4 15:43 fvSolution
-rw-r--r-- 1 gsclara gsclara  901 Mar  4 15:43 meshQualityDict
-rw-r--r-- 1 gsclara gsclara  177 Mar  4 15:43 profiling
-rw-r--r-- 1 gsclara gsclara 6630 Mar  4 15:43 runTimePostProcessing
-rw-r--r-- 1 gsclara gsclara  925 Mar  4 15:43 sampling
-rw-r--r-- 1 gsclara gsclara 2018 Mar  4 15:43 snappyHexMeshDict
-rw-r--r-- 1 gsclara gsclara  885 Mar  4 15:43 streamLines
-rw-r--r-- 1 gsclara gsclara  935 Mar  4 15:43 surfaceFeatureExtractDict
-rw-r--r-- 1 gsclara gsclara  403 Mar  4 15:43 visualization
-rw-r--r-- 1 gsclara gsclara 1961 Mar  4 15:43 vtkWrite
```

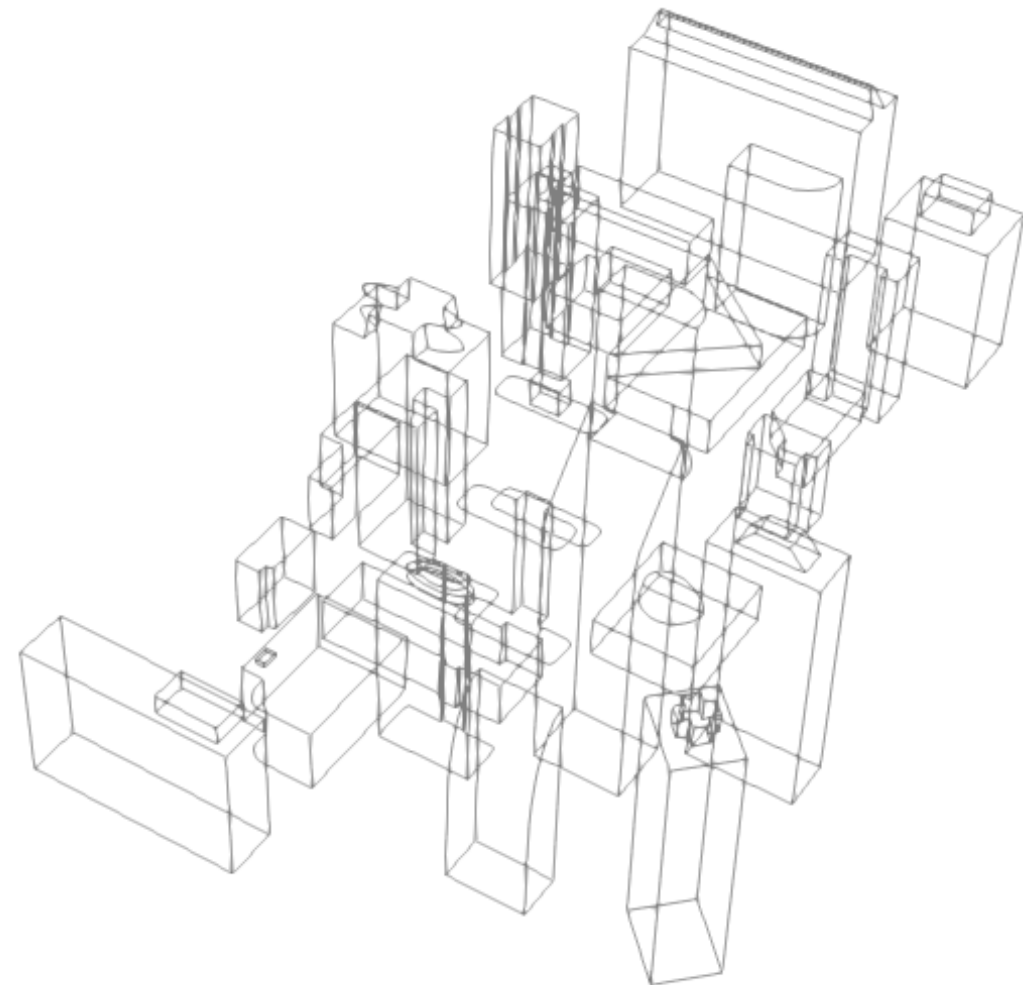
How to run openFOAM test case:

1. surfaceFeatures

2. blockMesh
3. snappyHexMesh
4. simpleFoam

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings/constant$ ls -la
total 28
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 15:46 .
drwxr-xr-x 9 gsclara gsclara 4096 Mar  4 15:46 ..
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 15:46 extendedFeatureEdgeMesh ←
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 15:46 polyMesh
-rw-r--r-- 1 gsclara gsclara  898 Mar  4 15:43 transportProperties
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 15:46 triSurface
-rw-r--r-- 1 gsclara gsclara  960 Mar  4 15:43 turbulenceProperties
```

With this command we can extract the edges, which can be used within snappyHexMesh, to help the snapping process and improve the quality of the mesh generation



How to run openFOAM test case:

1. surfaceFeatures
- 2. blockMesh**
3. snappyHexMesh
4. simpleFoam

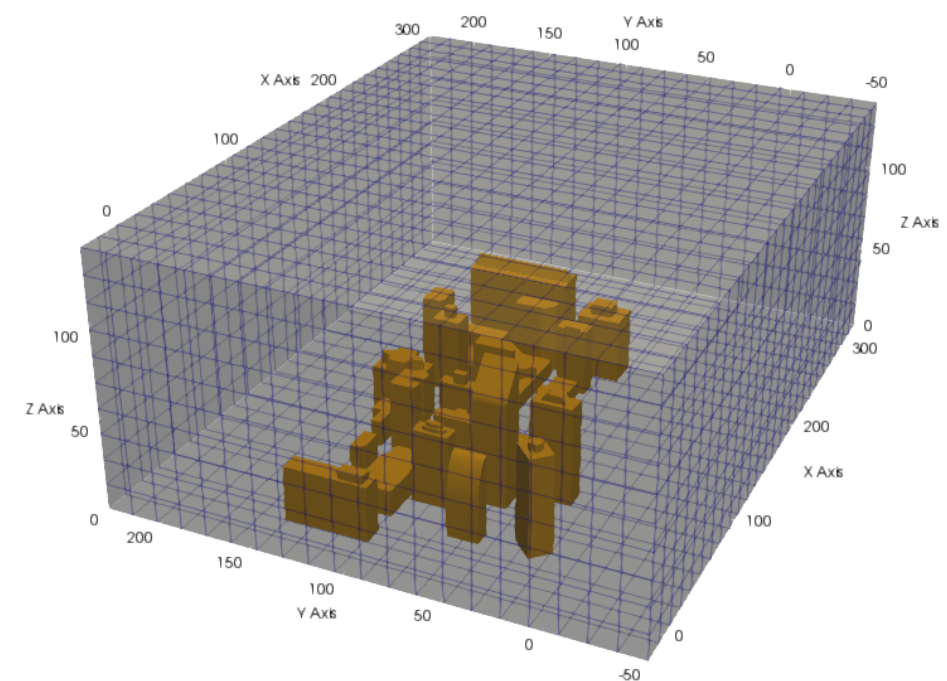
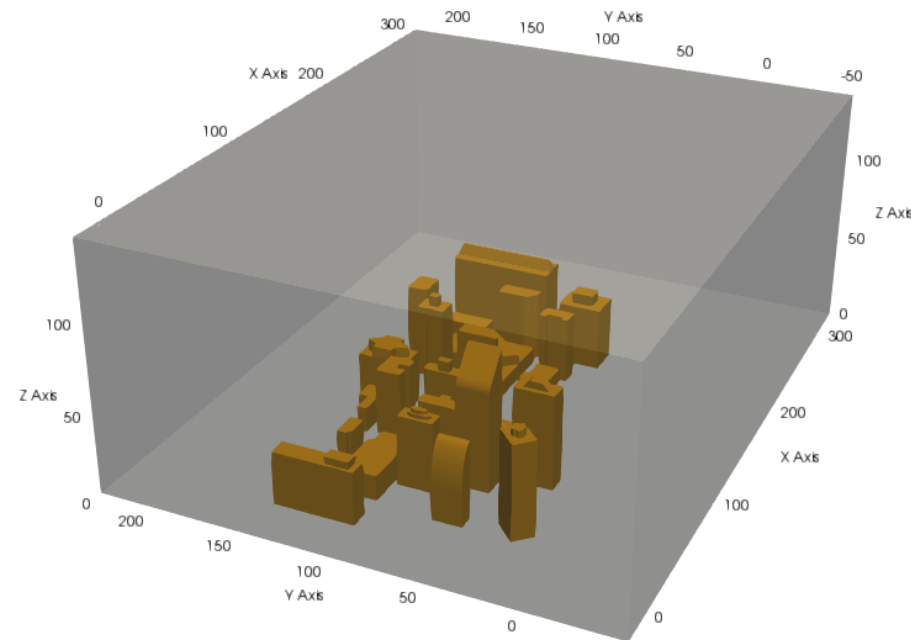
```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings$ ls -la constant/
total 28
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:12 .
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:11 ..
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:11 extendedFeatureEdgeMesh
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:12 polyMesh ←
-rw-r--r-- 1 gsclara gsclara  898 Mar  4 15:43 transportProperties
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:11 triSurface
-rw-r--r-- 1 gsclara gsclara  960 Mar  4 15:43 turbulenceProperties
```

```
// *****
4
(
  inlet
  {
    type            patch;
    nFaces          200;
    startFace       14050;
  }
  outlet
  {
    type            patch;
    nFaces          200;
    startFace       14250;
  }
  ground
  {
    type            wall;
    inGroups        List<word> 1(wall);
    nFaces          500;
    startFace       14450;
  }
  frontAndBack
  {
    type            symmetry;
    inGroups        List<word> 1(symmetry);
    nFaces          1000;
    startFace       14950;
  }
)
// *****
```

How to run openFOAM test case:

1. surfaceFeatures
- 2. blockMesh**
3. snappyHexMesh
4. simpleFoam

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings$ ls -la constant/
total 28
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:12 .
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:11 ..
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:11 extendedFeatureEdgeMesh
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:12 polyMesh ←
-rw-r--r-- 1 gsclara gsclara  898 Mar  4 15:43 transportProperties
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:11 triSurface
-rw-r--r-- 1 gsclara gsclara  960 Mar  4 15:43 turbulenceProperties
```

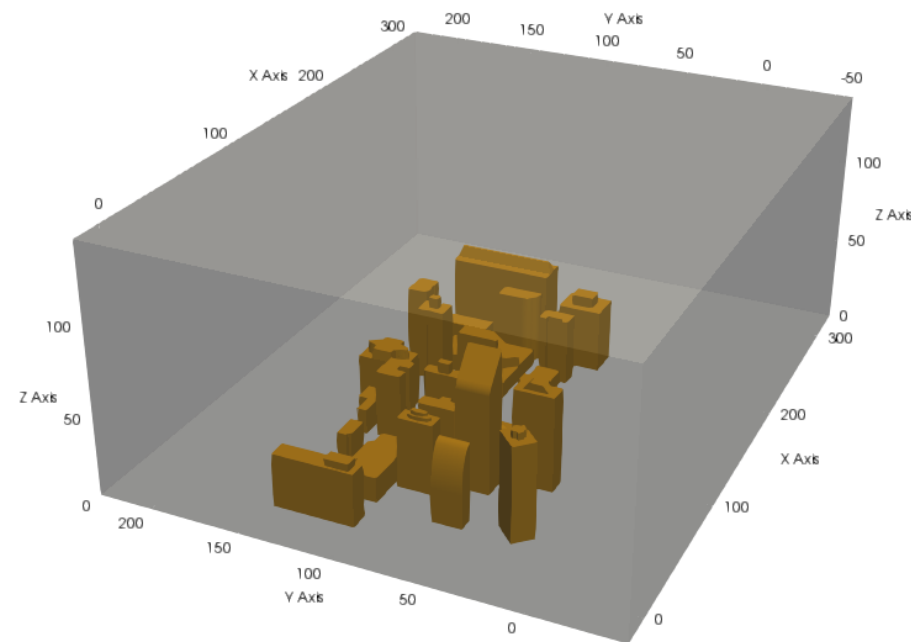


```
// *****
4
(
  inlet
  {
    type            patch;
    nFaces          200;
    startFace       14050;
  }
  outlet
  {
    type            patch;
    nFaces          200;
    startFace       14250;
  }
  ground
  {
    type            wall;
    inGroups        List<word> 1(wall);
    nFaces          500;
    startFace       14450;
  }
  frontAndBack
  {
    type            symmetry;
    inGroups        List<word> 1(symmetry);
    nFaces          1000;
    startFace       14950;
  }
)
// *****
```

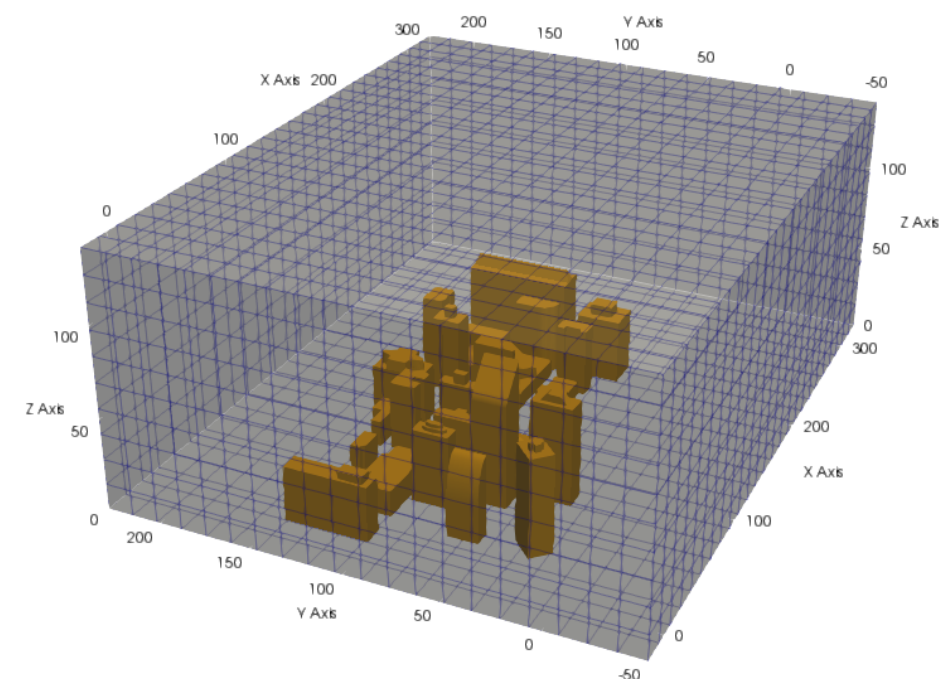
How to run openFOAM test case:

1. surfaceFeatures
- 2. blockMesh**
3. snappyHexMesh
4. simpleFoam

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings$ ls -la constant/
total 28
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:12 .
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:11 ..
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:11 extendedFeatureEdgeMesh
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:12 polyMesh
-rw-r--r-- 1 gsclara gsclara  898 Mar  4 15:43 transportProperties
drwxrwxr-x 2 gsclara gsclara 4096 Mar  4 16:11 triSurface
-rw-r--r-- 1 gsclara gsclara  960 Mar  4 15:43 turbulenceProperties
```



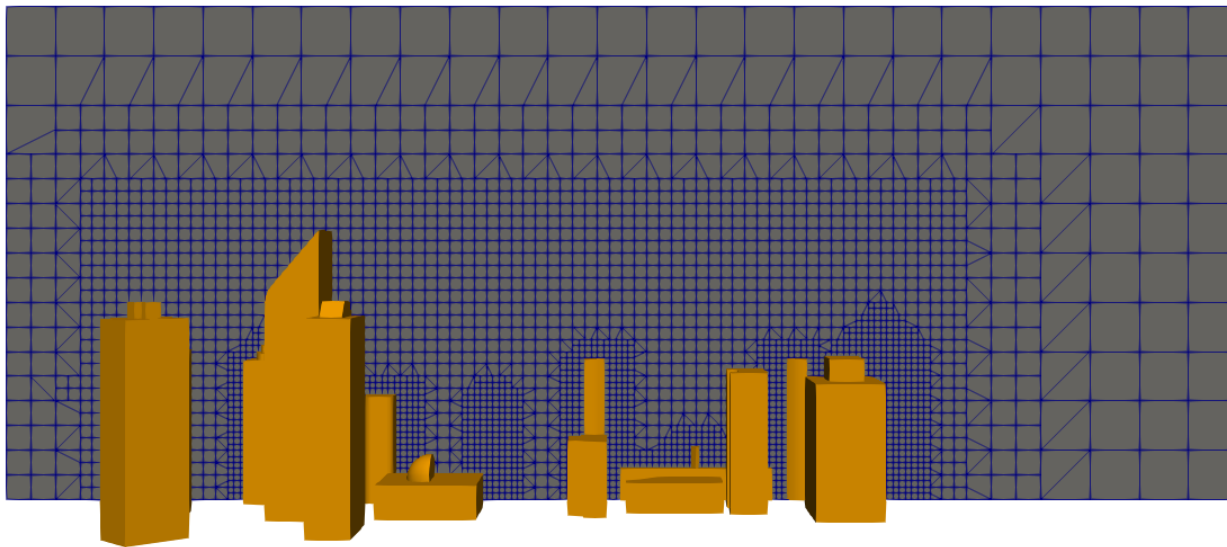
```
// *****
4
(
  inlet
  {
    type            patch;
    nFaces          200;
    startFace       14050;
  }
  outlet
  {
    type            patch;
    nFaces          200;
    startFace       14250;
  }
  ground
  {
    type            wall;
    inGroups        List<word> 1(wall);
    nFaces          500;
    startFace       14450;
  }
  frontAndBack
  {
    type            symmetry;
    inGroups        List<word> 1(symmetry);
    nFaces          1000;
    startFace       14950;
  }
)
// *****
```



How to run openFOAM test case:

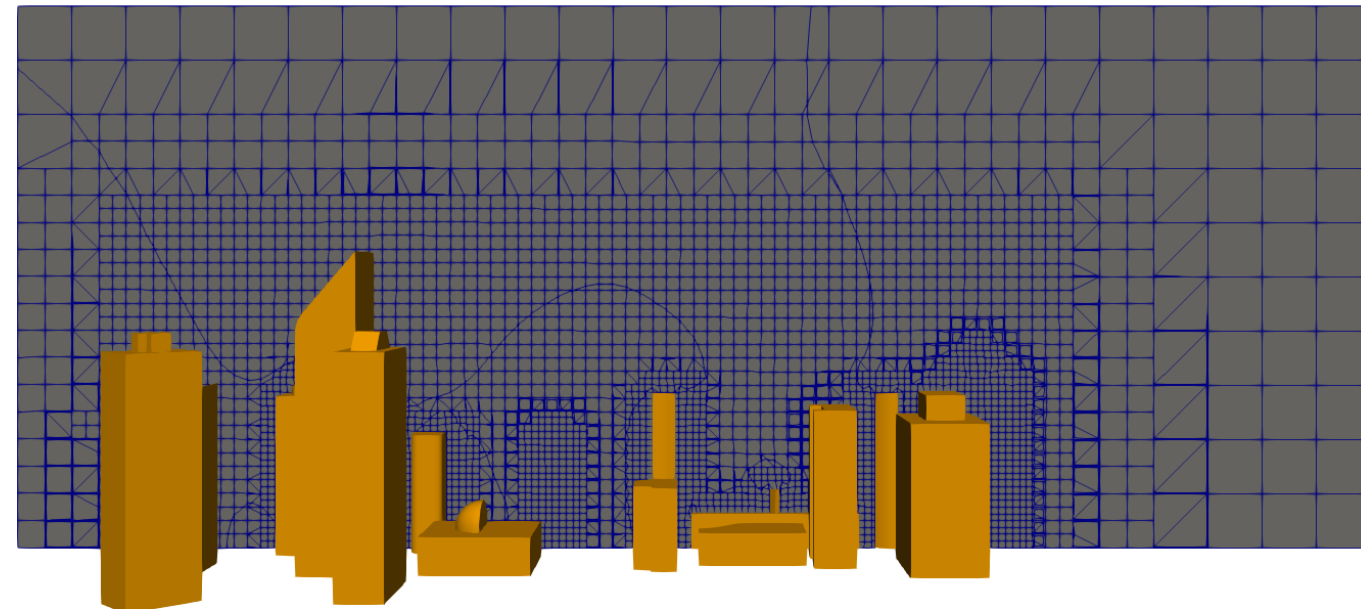
1. surfaceFeatures
2. blockMesh
- 3. snappyHexMesh**
4. simpleFoam

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings$ ls -la
total 60
drwxr-xr-x 7 gsclara gsclara 4096 Mar  4 16:13 .
drwxrwxr-x 7 gsclara gsclara 4096 Mar  4 15:43 ..
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:46 0.orig
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 1
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 2
-rwxr-xr-x 1 gsclara gsclara  408 Mar  4 15:43 Allclean
-rwxr-xr-x 1 gsclara gsclara  387 Mar  4 15:43 Allrun
-rwxr-xr-x 1 gsclara gsclara  516 Mar  4 15:43 Allrun-parallel
-rwxr-xr-x 1 gsclara gsclara  555 Mar  4 15:43 Allrun.pre
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:12 constant
-rw-rw-r-- 1 gsclara gsclara 2373 Mar  4 16:11 log.blockMesh
-rw-rw-r-- 1 gsclara gsclara 1667 Mar  4 16:11 log.snappyHexMesh
-rw-rw-r-- 1 gsclara gsclara 4443 Mar  4 16:11 log.surfaceFeatureExtract
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:58 system
```



```
#includeEtc "caseDicts/mesh/generation/snappyHexMeshDict.cfg"

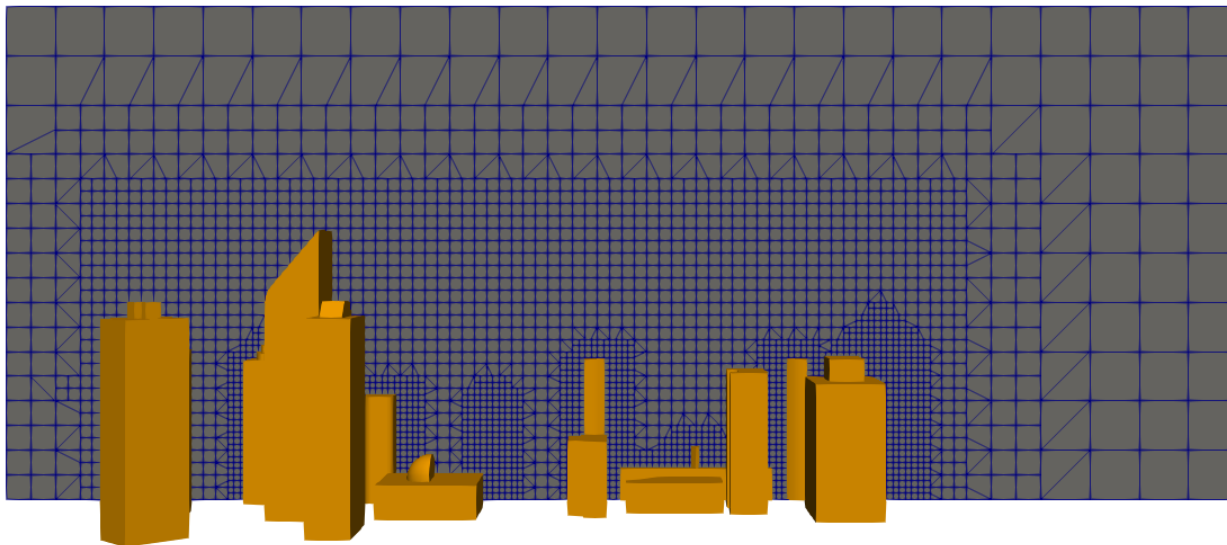
castellatedMesh on;
snap           on;
addLayers      off;
```



How to run openFOAM test case:

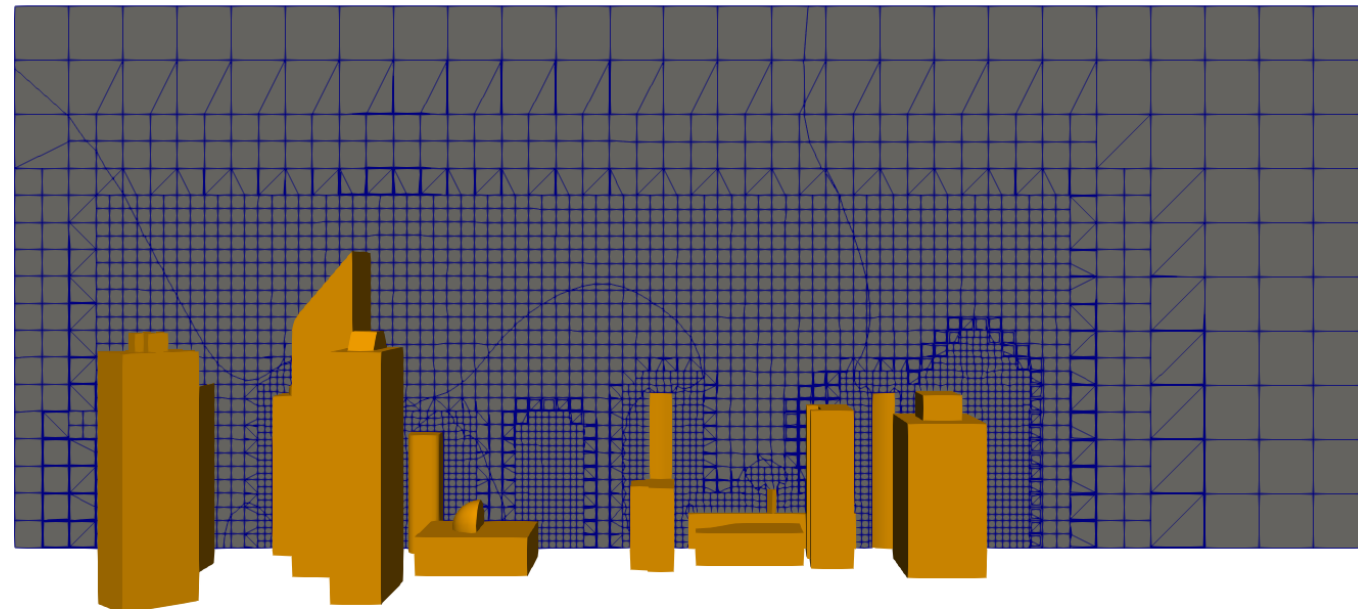
1. surfaceFeatures
2. blockMesh
- 3. snappyHexMesh**
4. simpleFoam

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings$ ls -la
total 60
drwxr-xr-x 7 gsclara gsclara 4096 Mar  4 16:13 .
drwxrwxr-x 7 gsclara gsclara 4096 Mar  4 15:43 ..
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:46 0.orig
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 1
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 2
-rwxr-xr-x 1 gsclara gsclara  408 Mar  4 15:43 Allclean
-rwxr-xr-x 1 gsclara gsclara  387 Mar  4 15:43 Allrun
-rwxr-xr-x 1 gsclara gsclara  516 Mar  4 15:43 Allrun-parallel
-rwxr-xr-x 1 gsclara gsclara  555 Mar  4 15:43 Allrun.pre
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:12 constant
-rw-rw-r-- 1 gsclara gsclara 2373 Mar  4 16:11 log.blockMesh
-rw-rw-r-- 1 gsclara gsclara 1667 Mar  4 16:11 log.snappyHexMesh
-rw-rw-r-- 1 gsclara gsclara 4443 Mar  4 16:11 log.surfaceFeatureExtract
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:58 system
```



```
#includeEtc "caseDicts/mesh/generation/snappyHexMeshDict.cfg"

1. castellatedMesh on;
2. snap          on;
   addLayers     off;
```

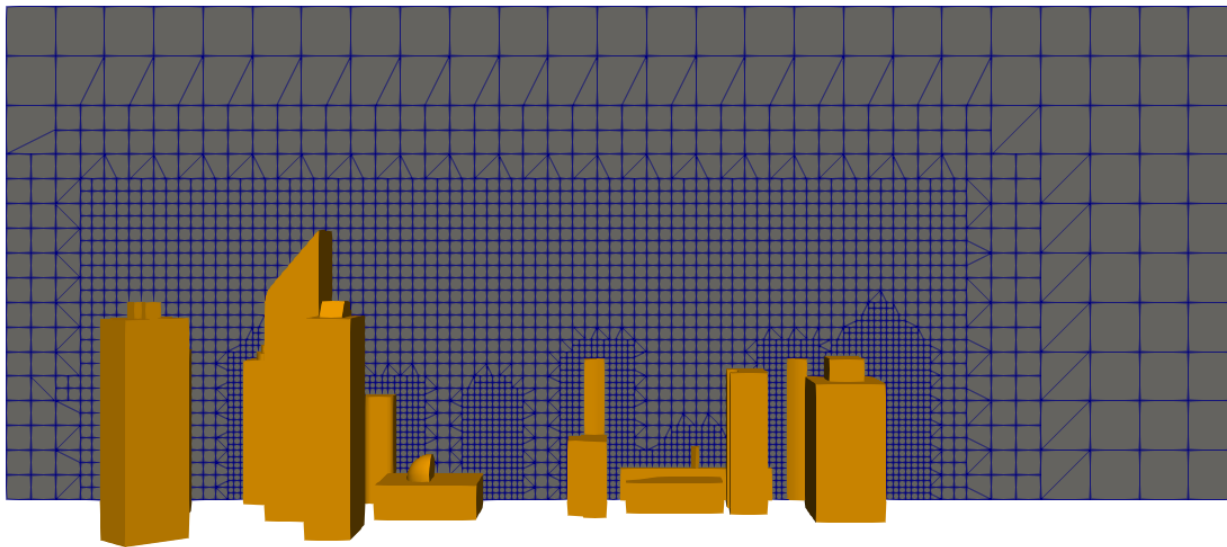


How to run openFOAM test case:

1. surfaceFeatures
2. blockMesh
- 3. snappyHexMesh**
4. simpleFoam

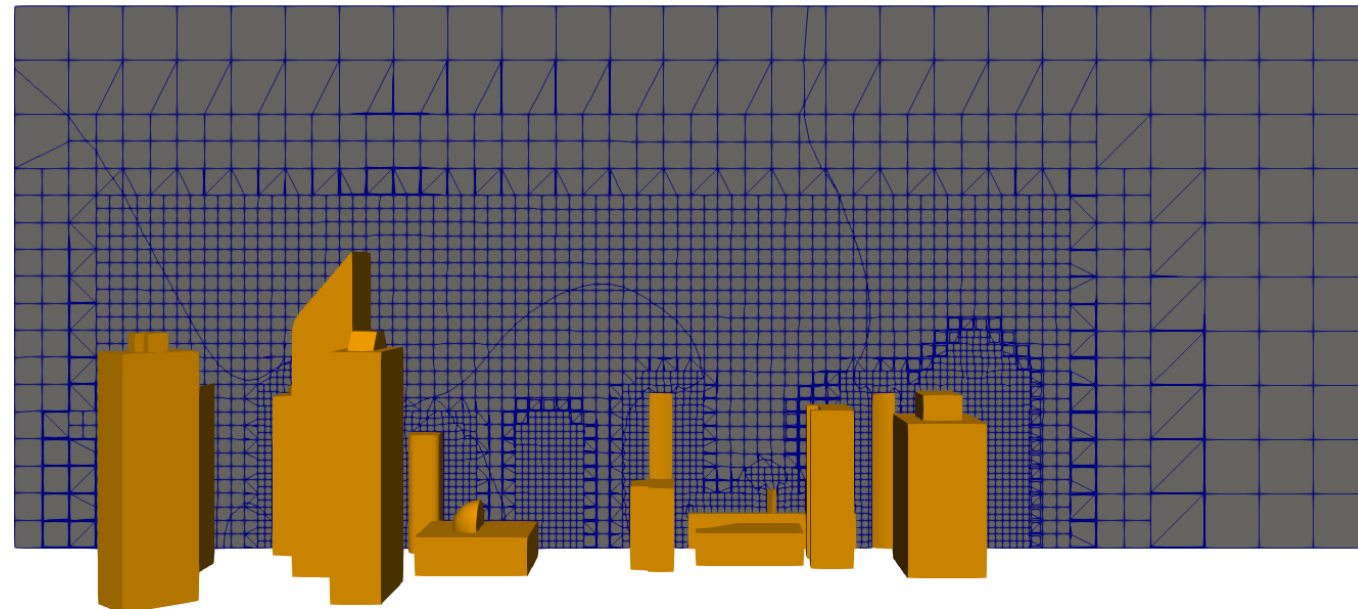
```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings$ ls -la
total 60
drwxr-xr-x 7 gsclara gsclara 4096 Mar  4 16:13 .
drwxrwxr-x 7 gsclara gsclara 4096 Mar  4 15:43 ..
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:46 0.orig
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 1
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 2
-rwxr-xr-x 1 gsclara gsclara  408 Mar  4 15:43 Allclean
-rwxr-xr-x 1 gsclara gsclara  387 Mar  4 15:43 Allrun
-rwxr-xr-x 1 gsclara gsclara  516 Mar  4 15:43 Allrun-parallel
-rwxr-xr-x 1 gsclara gsclara  555 Mar  4 15:43 Allrun.pre
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:12 constant
-rw-rw-r-- 1 gsclara gsclara 2373 Mar  4 16:11 log.blockMesh
-rw-rw-r-- 1 gsclara gsclara 1667 Mar  4 16:11 log.snappyHexMesh
-rw-rw-r-- 1 gsclara gsclara 4443 Mar  4 16:11 log.surfaceFeatureExtract
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:58 system
```

1. Castellation



```
#includeEtc "caseDicts/mesh/generation/snappyHexMeshDict.cfg"

1. castellatedMesh on;
2. snap          on;
   addLayers      off;
```

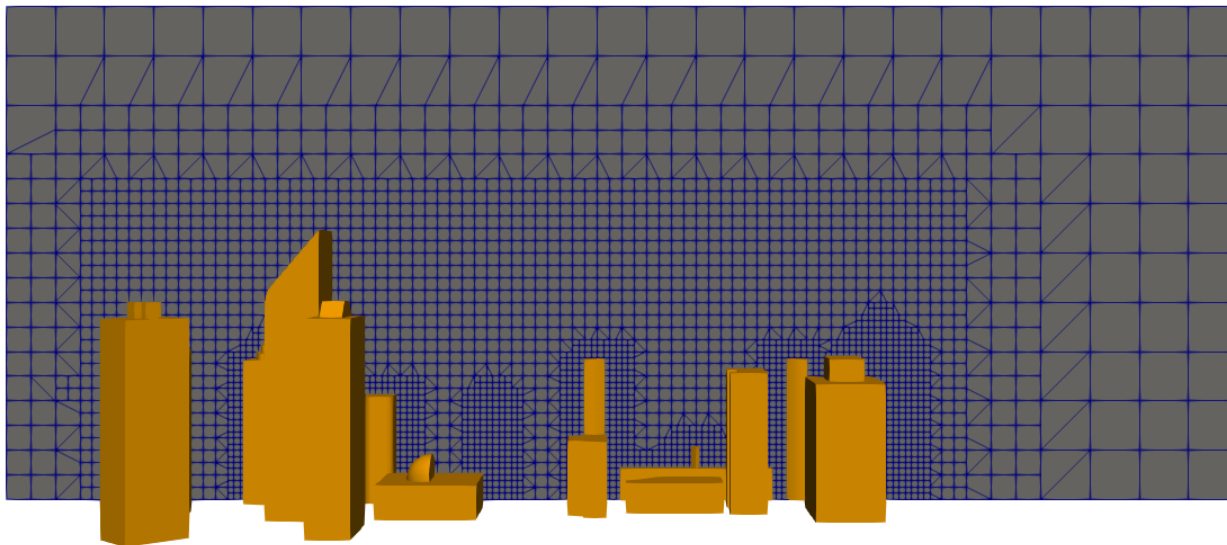


How to run openFOAM test case:

1. surfaceFeatures
2. blockMesh
- 3. snappyHexMesh**
4. simpleFoam

```
gsclara@foote:~/OpenFOAM/gsclara-v2412/run/windAroundBuildings$ ls -la
total 60
drwxr-xr-x 7 gsclara gsclara 4096 Mar  4 16:13 .
drwxrwxr-x 7 gsclara gsclara 4096 Mar  4 15:43 ..
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:46 0.orig
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 1
drwxrwxr-x 3 gsclara gsclara 4096 Mar  4 16:13 2
-rwxr-xr-x 1 gsclara gsclara  408 Mar  4 15:43 Allclean
-rwxr-xr-x 1 gsclara gsclara  387 Mar  4 15:43 Allrun
-rwxr-xr-x 1 gsclara gsclara  516 Mar  4 15:43 Allrun-parallel
-rwxr-xr-x 1 gsclara gsclara  555 Mar  4 15:43 Allrun.pre
drwxr-xr-x 5 gsclara gsclara 4096 Mar  4 16:12 constant
-rw-rw-r-- 1 gsclara gsclara 2373 Mar  4 16:11 log.blockMesh
-rw-rw-r-- 1 gsclara gsclara 1667 Mar  4 16:11 log.snappyHexMesh
-rw-rw-r-- 1 gsclara gsclara 4443 Mar  4 16:11 log.surfaceFeatureExtract
drwxr-xr-x 2 gsclara gsclara 4096 Mar  4 15:58 system
```

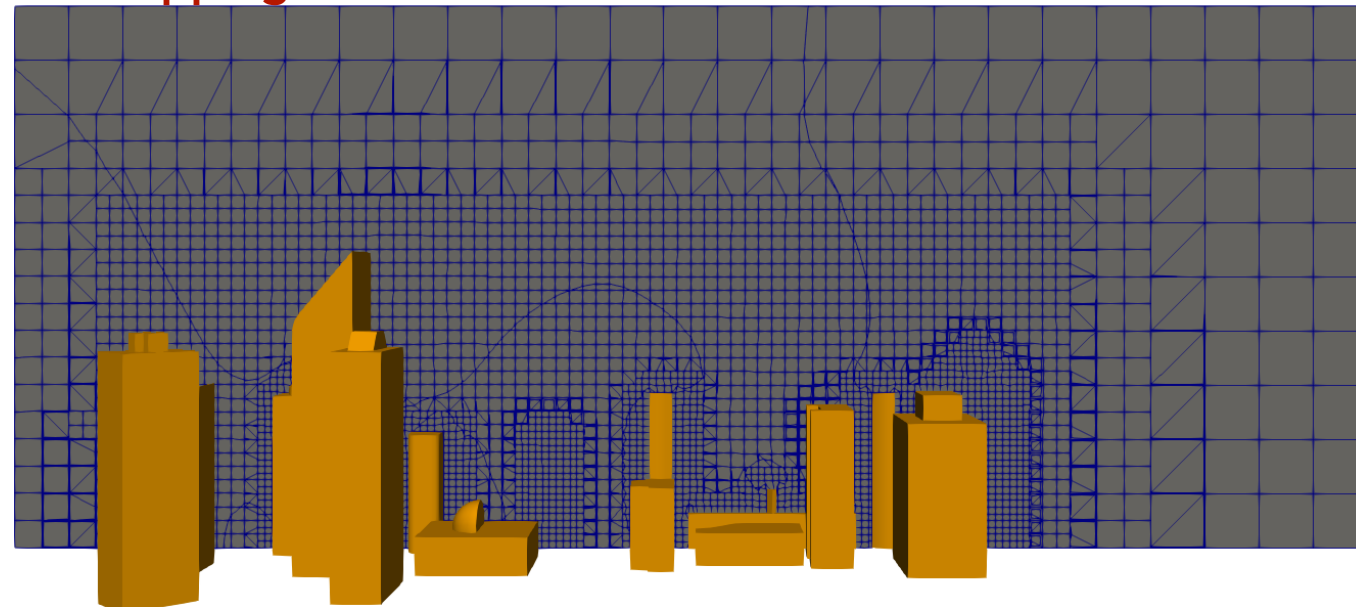
1. Castellation



```
#includeEtc "caseDicts/mesh/generation/snappyHexMeshDict.cfg"

1. castellatedMesh on;
2. snap          on;
   addLayers      off;
```

2. Snapping



How to run openFOAM test case:

1. surfaceFeatures
2. blockMesh
3. snappyHexMesh
4. **simpleFoam**

```
SIMPLE
{
    residualControl
    {
        p            1e-4;
        U            1e-4;
        "(k|omega|epsilon)" 1e-4;
    }
    nNonOrthogonalCorrectors 0;
    pRefCell          0;
    pRefValue          0;
}
```

Time = 399

```
smoothSolver: Solving for Ux, Initial residual = 0.000374315, Final residual = 2.13748e-05, No Iterations 2
smoothSolver: Solving for Uy, Initial residual = 0.00149301, Final residual = 8.74602e-05, No Iterations 2
smoothSolver: Solving for Uz, Initial residual = 0.00168795, Final residual = 0.000102205, No Iterations 2
GAMG: Solving for p, Initial residual = 0.0132326, Final residual = 0.000431114, No Iterations 2
time step continuity errors : sum local = 1.12228e-05, global = 1.23819e-06, cumulative = 0.00416481
smoothSolver: Solving for epsilon, Initial residual = 0.00286634, Final residual = 0.000159461, No Iterations 1
smoothSolver: Solving for k, Initial residual = 0.00206614, Final residual = 0.000165122, No Iterations 1
ExecutionTime = 241.25 s  ClockTime = 247 s
```

Time = 400

```
smoothSolver: Solving for Ux, Initial residual = 0.000373325, Final residual = 2.13718e-05, No Iterations 2
smoothSolver: Solving for Uy, Initial residual = 0.00148155, Final residual = 8.70915e-05, No Iterations 2
smoothSolver: Solving for Uz, Initial residual = 0.00167385, Final residual = 0.000101258, No Iterations 2
GAMG: Solving for p, Initial residual = 0.0132477, Final residual = 0.000427785, No Iterations 2
time step continuity errors : sum local = 1.11348e-05, global = 1.15567e-06, cumulative = 0.00416596
smoothSolver: Solving for epsilon, Initial residual = 0.00288552, Final residual = 0.000160044, No Iterations 1
smoothSolver: Solving for k, Initial residual = 0.00205397, Final residual = 0.000165605, No Iterations 1
ExecutionTime = 241.83 s  ClockTime = 248 s
```

How to run openFOAM test case:

1. surfaceFeatures
2. blockMesh
3. snappyHexMesh
4. **simpleFoam**

```
SIMPLE
{
    residualControl
    {
        p            1e-4;
        U            1e-4;
        "(k|omega|epsilon)" 1e-4;
    }
    nNonOrthogonalCorrectors 0;
    pRefCell          0;
    pRefValue          0;
}
```

```
// *****
application      simpleFoam;
startFrom         latestTime;
startTime         0;
stopAt            endTime;
endTime           400;
deltaT            1;
writeControl       timeStep;
writeInterval      50;
```

```
Time = 399
smoothSolver: Solving for Ux, Initial residual = 0.000374315, Final residual = 2.13748e-05, No Iterations 2
smoothSolver: Solving for Uy, Initial residual = 0.00149301, Final residual = 8.74602e-05, No Iterations 2
smoothSolver: Solving for Uz, Initial residual = 0.00168795, Final residual = 0.000102205, No Iterations 2
GAMG: Solving for p, Initial residual = 0.0132326, Final residual = 0.000431114, No Iterations 2
time step continuity errors : sum local = 1.12228e-05, global = 1.23819e-06, cumulative = 0.00416481
smoothSolver: Solving for epsilon, Initial residual = 0.00286634, Final residual = 0.000159461, No Iterations 1
smoothSolver: Solving for k, Initial residual = 0.00206614, Final residual = 0.000165122, No Iterations 1
ExecutionTime = 241.25 s  ClockTime = 247 s

Time = 400
smoothSolver: Solving for Ux, Initial residual = 0.000373325, Final residual = 2.13718e-05, No Iterations 2
smoothSolver: Solving for Uy, Initial residual = 0.00148155, Final residual = 8.70915e-05, No Iterations 2
smoothSolver: Solving for Uz, Initial residual = 0.00167385, Final residual = 0.000101258, No Iterations 2
GAMG: Solving for p, Initial residual = 0.0132477, Final residual = 0.000427785, No Iterations 2
time step continuity errors : sum local = 1.11348e-05, global = 1.15567e-06, cumulative = 0.00416596
smoothSolver: Solving for epsilon, Initial residual = 0.00288552, Final residual = 0.000160044, No Iterations 1
smoothSolver: Solving for k, Initial residual = 0.00205397, Final residual = 0.000165605, No Iterations 1
ExecutionTime = 241.83 s  ClockTime = 248 s
```

How to run openFOAM test case:

1. surfaceFeatures
2. blockMesh
3. snappyHexMesh

4. simpleFoam IN PARALLEL

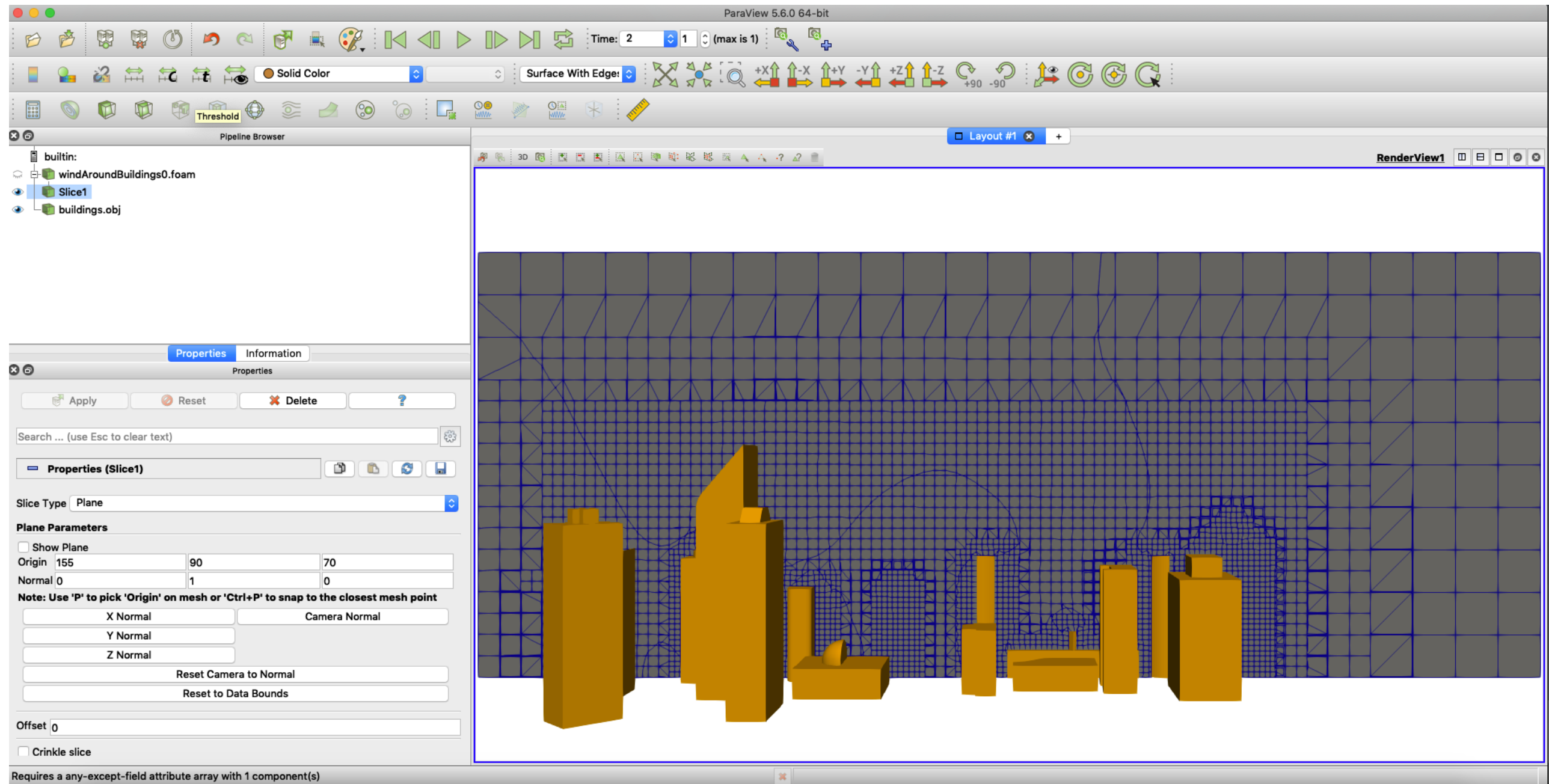
```
cp $FOAM_ETC/caseDicts/preProcessing/decomposeParDict system/
```

```
mpirun -np 4 simpleFoam -parallel >& log.SF &
```

```
tail -f log.SF
```

```
reconstructPar -latestTime
```

■ Paraview:



<https://www.paraview.org/Wiki/images/6/6e/ParaViewTutorial54.pdf>

Activities before next class

- Go to: <https://www.paraview.org/Wiki/images/6/6e/ParaViewTutorial54.pdf>
- Make the different plots suggested in the tutorial
- Run the case switching off turbulence
- Run the case increasing the **ν** (viscosity) to honey level (2000) or oil level (56.2)
- What happens if you change endTime in controlDict